

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)		Field lines radial minimum of 3 (1) Field lines / arrows towards the Earth (1) Equipotentials are spheres minimum of 1 (circular cross-section) (1)	3			3		
	(b)	(i)	Use of $V = [-]\frac{GM}{r}$ (1) Answer = 1.16×10^6 or 1.29×10^5 (1) Answer = -1.29×10^6 *** J kg ⁻¹ *** unit mark (1)	1	1 1		3	1	
		(ii)	Use of $g = [-]\frac{GM}{r^2}$ (implied by 0.0034) (1) Evidence of cancelling fields e.g. $\frac{597}{342^2} - \frac{7.37}{38^2}$ or $0.0034 - 0.0034$ or difference in fields = 1.71×10^{-7} or checking ratio of fields is equal to 1 (1)	1	1		2	1	
		(iii)	Definition of SHM or force / acceleration proportional [& opposite] to displacement (can be implied through logical argument) (1) Force is in same direction as displacement / space craft will accelerate towards Moon or Earth (1) Hence, Dafydd is wrong consistent with argument and ecf (1)			3	3		
			Question 2 total	5	3	3	11	2	0